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ENCAPSULATED MEMS DEVICES

Abstract

In examples, a micro-electromechanical (MEMS) device comprises a substrate and a semiconductor die coupled to the substrate and including circuitry formed therein. The semiconductor die also includes a bond pad coupled to the circuitry. The MEMS device includes a structure extending away from the semiconductor die and having four sides, with the structure comprising a corrodible material. The MEMS device includes an epoxy contacting outer surfaces of the four sides of the structure and over the corrodible material.

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Background/Summary

BACKGROUND

[0001] Micro-electromechanical systems (MEMS) devices integrate miniature mechanical elements, sensors, actuators, and electronics onto a single silicon substrate. These compact systems, typically ranging from micrometers to millimeters in size, use microfabrication principles to create functional devices. MEMS technology has diverse applications across industries such as aerospace, healthcare, telecommunications, and consumer electronics. The miniaturization and integration capabilities of MEMS devices enable the development of efficient and cost-effective solutions, offering improved performance and functionality compared to traditional, macro-scale counterparts.

SUMMARY

[0002] In examples, a micro-electromechanical (MEMS) device includes a substrate and a semiconductor die coupled to the substrate and including circuitry formed therein. The semiconductor die also includes a bond pad coupled to the circuitry. The MEMS device includes a structure extending away from the semiconductor die and having four sides, with the structure including a corrodible material. The MEMS device includes an epoxy contacting outer surfaces of the four sides of the structure and over the corrodible material.

[0003] In examples, a method of manufacturing a micro-electromechanical (MEMS) device includes attaching a first semiconductor die to a substrate panel and attaching a first structure to the substrate panel around the first semiconductor die, with the first structure including a first metal stack. The method includes attaching a second semiconductor die to the substrate panel and attaching a second structure to the substrate panel around the second semiconductor die, with the second structure including a second metal stack. The method includes depositing an epoxy contacting first and second surfaces of the first structure and contacting third and fourth surfaces of the second structure and in a gap between the second surface of the first structure and the third surface of the second structure. The method includes singulating the substrate panel including cutting through epoxy in the gap and through a portion of the substrate panel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1**A is a block diagram of an automotive system including an encapsulated MEMS device, in accordance with various examples.

[0005] FIG. **1**B is a block diagram of a system including an encapsulated MEMS device, in accordance with various examples.

[0006] FIGS. **2**A and **2**B are different cross-sectional views of an encapsulated MEMS device, in accordance with various examples.

[0007] FIG. **2**C is a top-down view of an encapsulated MEMS device, in accordance with various examples.

[0008] FIG. **2**D is a perspective view of an encapsulated MEMS device, in accordance with various examples.

[0009] FIG. **2**E is a cross-sectional view of an encapsulated MEMS device, in accordance with various examples.

[0010] FIG. **2**F is a top-down view of an encapsulated MEMS device, in accordance with various examples.

[0011] FIG. **2**G is a perspective view of an encapsulated MEMS device, in accordance with various examples.

[0012] FIGS. **3**A and **3**B are different cross-sectional views of an encapsulated MEMS device, in accordance with various examples.

[0013] FIG. **3**C is a top-down view of an encapsulated MEMS device, in accordance with various examples.

[0014] FIG. **3**D is a perspective view of an encapsulated MEMS device, in accordance with various examples.

[0015] FIG. **3**E is a cross-sectional view of an encapsulated MEMS device, in accordance with various examples.

[0016] FIG. **3**F is a top-down view of an encapsulated MEMS device, in accordance with various examples.

[0017] FIG. **3**G is a perspective view of an encapsulated MEMS device, in accordance with various examples.

[0018] FIG. **4** is a flow diagram of a method for manufacturing an encapsulated MEMS device, in accordance with various examples.

[0019] FIGS. **5**A**1**-**5**E**3** are a process flow for manufacturing an encapsulated MEMS device, in accordance with various examples.

[0020] FIG. **6** is a flow diagram of a method for manufacturing an encapsulated MEMS device, in accordance with various examples.

[0021] FIGS. **7**A**1**-**7**E**3** are a process flow for manufacturing an encapsulated MEMS device, in accordance with various examples.

DETAILED DESCRIPTION

[0022] MEMS devices have varying structures and operations. For example, some MEMS devices include a substrate, a semiconductor die containing circuitry, an array of mirrors at a device side of the die, and a glass panel above the array of mirrors. Such MEMS devices are particularly useful in optical applications. For instance, the mirrors may be configured to reflect incoming light in particular directions to achieve application-specific objectives. Other MEMS devices may be structured differently, containing accelerometers, gyroscopes, sensors, and other such micro-scale components that are useful in a variety of other applications. Many types of MEMS devices include orifices through which deleterious environmental influences, such as moisture and debris, can enter and damage vulnerable components, such as metal stacks, glass panels, and mirror arrays.

[0023] This disclosure describes various examples of a MEMS device that mitigates the risk of damage by environmental factors such as moisture and debris, as described above. In examples, a MEMS device includes a substrate having a metal trace exposed to an outer surface of the substrate. The MEMS device includes a semiconductor die coupled to the substrate and a bond pad coupled to the metal trace by way of a bond wire. The semiconductor die has a device side including a mirror. The MEMS device further includes a four-sided structure including a metal stack. The structure is coupled to the device side of the semiconductor die and to a glass member. The structure forms a sealed cavity between the semiconductor die and the glass member. The MEMS device includes an epoxy contacting each of the four surfaces of the structure. The epoxy seals orifices between the semiconductor die and the glass member, protecting vulnerable components of the MEMS device, such as the aforementioned structure (which may be composed of, e.g., one or more corrodible metals). Consequently, the example MEMS devices described herein experience substantially increased longevity, with the example MEMS devices gaining at least an additional 10,864 hours of operational life over existing MEMS devices in experimental salt challenge testing.

[0024] FIG. **1**A is a block diagram of an automotive system **100** including an encapsulated MEMS device, in accordance with various examples. The automotive system **100** may be a car, pickup truck, cargo truck, etc. The automotive system **100** may include a MEMS device **104**, which may take the form of any of the various example MEMS devices described herein. Within the automotive system **100**, the MEMS device **104** may be deployed in a variety of applications, including displays, projection systems, airbag systems, tire pressure monitoring systems, fuel injection systems, navigation systems, electronic stability control systems, engine control systems, etc.

[0025] FIG. **1**B is a block diagram of a system **150** including an encapsulated MEMS device, in accordance with various examples. The system **150** may be any type of electronic device, such as automobiles, electronic appliances (e.g., refrigerators, microwave ovens, clothing washers and dryers), media devices (e.g., projectors, televisions, audio equipment), cameras, smartphones, personal computers, notebooks, tablets, aircraft, spacecraft, etc. The system **150** also may find application in marine environments, and in at least some examples, may take the form of watercraft. The system **150** may include a MEMS device **104**, which may take the form of any of the various example MEMS devices described herein.
[0026] FIGS. **2**A and **2**B are different cross-sectional views of an encapsulated MEMS device, in accordance with various examples. FIG. **2**C is a top-down view (i.e., a view toward the active circuitry of the MEMS device, as opposed to a non-device side of the MEMS device) of the MEMS device of FIGS. **2**A and **2**B, in accordance with various examples. FIG. **2**D is a perspective view of the MEMS device illustrated in FIGS. **2**A-**2**C, in accordance with various examples. The cross-section of FIG. **2**A is taken along line **290** shown in FIG. **2**C, and the cross-section of FIG. **2**B is taken along line **292** shown in FIG. **2**C.
[0027] More particularly, FIGS. **2**A-**2**D show a MEMS device **250**, which is an example of the MEMS device **104** described above. The MEMS device **250** includes a substrate **200**, which in some examples may be a ceramic substrate. The MEMS device **250** includes a semiconductor die **202** on the substrate **200**. The semiconductor die **202** may include one or more devices **204** on a device side **205** of the semiconductor die **202**. The one or more devices **204** may include circuitry (e.g., complementary metal oxide semiconductor (CMOS) circuitry), one or more mirrors, other MEMS mechanical devices, etc. that may be susceptible to damage by moisture, salt, debris, and other external contaminants. The MEMS device **250** includes bond pads **206** and bond leads **208** on the substrate **200**. A ball bond **210** is coupled to the bond pad **206**, and a bond wire **212** is coupled to the ball bond **210** and to the bond lead **208** (e.g., such as by a stitch bond). Other types of bonds besides ball bonds and stitch bonds are contemplated and included in the scope of this disclosure.
[0028] In examples, the MEMS device **250** includes a cavity **240** in which the semiconductor die **202** is positioned. However, in other examples, the cavity **240** may be omitted, and the semiconductor die **202** may be placed on a flat surface that is horizontally coplanar with other surfaces within the MEMS device **250**, such as a surface **242** or a shelf **244**. In examples, the bond leads **208** are positioned on the shelf **244**, which circumscribes the semiconductor die **202**. However, in examples, the shelf **244** may be omitted, and the bond leads **208** may be positioned elsewhere, such as on the surface **242**.
[0029] An epoxy **214** covers various components of the MEMS device **250**, such as the bond pads **206**, bond leads **208**, the ball bond **210**, the bond wire **212**, and portions of the substrate **200**. The epoxy **214** contacts and partially covers, but does not completely surround, the substrate **200**. The epoxy **214** also contacts and covers a multi-sided (e.g., four-sided) structure **216** that extends approximately orthogonally from the device side **205** in the vertical direction and extends approximately parallel to each side of a perimeter of the semiconductor die **202** in the horizontal direction. In examples, the epoxy **214** contacts and covers the outer surfaces of the multi-sided structure **216**. The epoxy **214** contacts and partially covers a cap **218** (e.g., a glass panel; a filter for gas or liquid sensing applications; fluidic channels in microfluidic applications) that is coupled to the structure **216**. The cap **218** is approximately parallel to the semiconductor die **202**. The structure **216** may be composed of a corrodible material. The structure **216** may include, for instance, a metal stack and an optional interposer (e.g., a semiconductor interposer) that, together with the cap **218**, form a seal (e.g., a hermetic seal) enclosing a cavity **220**. The epoxy **214** contacts and covers an outer surface **221** of the structure **216** that faces away from the cavity **220**. The epoxy **214** contacts and covers part of the cap **218**, but in examples, the epoxy **214** does not contact or cover any portion of a top surface **223** of the cap **218**. A gap **201** between the substrate **200** and the semiconductor die **202** is sufficiently large that the semiconductor die **202** may be placed in the

substrate **200** without colliding with one or more walls of the substrate **200**. The epoxy **214** seals the various components that the epoxy **214** contacts and covers and is fluid-resistant, protecting such components from moisture, debris, and other damaging environmental influences.

[0030] The epoxy **214** may be a glob top epoxy, a mold compound, or any other suitable epoxy or non-epoxy material that serves the purposes and performs the functions attributed herein to the epoxy **214**. The epoxy **214** is composed of at least 80% silica, because a composition lower than 80% silica presents technical disadvantages such as the application of mechanical and/or thermal stress to the MEMS device **250** that can damage various components of the MEMS device **250** (e.g., cracking of the cap **218**, breaking and/or lifting of bond wires **212**, cracking of the substrate **200**). The epoxy **214** has a coefficient of thermal expansion below 25, because a coefficient of thermal expansion at or above 25 presents technical disadvantages, such as the application of mechanical and/or thermal stress to the MEMS device **250** that can damage various components of the MEMS device **250** (e.g., cracking of the cap **218**, breaking and/or lifting of bond wires **212**, cracking of the substrate **200**). The epoxy **214** has a thickness adequate to cover all metals, alloys, and oxide layers in the structure **216**, as well as the bond wires **212**. Covering the structure **216** includes covering most or all of the surface **221** (e.g., including any orifices, interfaces between layers, and metal surfaces), as well as the interface between the structure **216** and the cap **218**. A thickness of the epoxy **214** that fails to cover most or all (e.g., including any orifices, interfaces between layers, and metal surfaces) of the surface **221** as well as the interface between the structure **216** and the cap **218** is technically disadvantageous because the structure **216**, and particularly the metals in the structure **216**, is vulnerable to corrosion and/or oxidation, and because the MEMS device **250** becomes vulnerable to infiltration and damage by external contaminants, such as salt, moisture, etc. However, if the epoxy **214** is so thick that the epoxy **214** covers substantially more than the surface **221** and the interface between the structure **216** and the cap **218**, this thickness becomes technically disadvantageous by adding expense and bulk without commensurate benefit, and by possibly covering some or all of the top surface of the cap **218** in optical applications (e.g., if the cap **218** is a glass panel). Further, an excess of epoxy **214** in the lateral direction (i.e., covering more of the substrate **200** than necessary) can cause challenges in deploying the MEMS device **250** in certain systems, such as those systems that use parts of the substrate **200** as optical reference points to ensure that the MEMS device **250** is properly seated and aligned within the system.

[0031] The substrate **200** includes a metal trace **222** that is coupled to the bond lead **208** and to a metal contact **224** on an exterior and/or bottom surface **225** of the substrate **200**. The metal trace **222**, the bond lead **208**, the bond wire **212**, the ball bond **210**, and the bond pad **206** establish an electrical pathway between the metal contact **224** and circuitry of the semiconductor die **202**.

[0032] The view of FIG. **2**B is similar to the view of FIG. **2**A, except that in the cross-section of FIG. **2**B, bond pads and bond wires are not present. Despite the absence of bond pads and bond wires in the cross-section of FIG. **2**B, the epoxy **214** is still present, contacting and covering some or all of the substrate **200**, the semiconductor die **202**, the structure **216**, and the cap **218**.

[0033] As shown and as described above, the structure **216** is four-sided, with each of the four sides extending approximately parallel to a different side of the semiconductor die **202**. The four sides of the structure **216** thus intersect approximately at right angles with each other. The epoxy **214** contacts and covers the outer surfaces **221** of the four sides of the structure **216**.

[0034] The manner in which a MEMS device is manufactured (example manufacturing techniques are described in detail below) can affect the configuration of the epoxy **214**. In the example MEMS device **250** illustrated in FIGS. **2**A-**2**D, the epoxy **214** extends in the horizontal direction from the structure **216** and cap **218** to the substrate **200**, but the epoxy **214** does not extend to the outer perimeter of the substrate **200**. In contrast, as shown in the cross-sectional, top-down, and perspective views of FIGS. **2**E-**2**G, respectively, an example MEMS device **252** (which is an example of MEMS device **104**) may include an epoxy **215** extends in the horizontal direction from the structure **216** and the cap **218** to the outer perimeter of the substrate **200** (e.g., to two sides of the outer perimeter of the substrate **200**, or to four sides of the outer perimeter of the substrate **200**). Line **254** in FIG. **2**F indicates the cross-section of FIG. **2**E. The greater area occupied by the epoxy **215** in the example MEMS device **252** of FIG. **2**E and the lesser area occupied by the epoxy **214** in the example MEMS device **250** of FIGS. **2**A-**2**D is a function of the specific manufacturing technique used to form the MEMS device, and, as mentioned, the different manufacturing techniques resulting in these differing epoxy configurations are described in detail below.

[0035] FIGS. **3**A and **3**B are different cross-sectional views of an example MEMS device **350** (which is an example of MEMS device **104**), in accordance with various examples. FIG. **3**C is a top-down view of the MEMS device **350** of FIGS. **3**A and **3**B, in accordance with various examples. FIG. **3**D is a perspective view of the MEMS device **350** of FIGS. **3**A-**3**C, in accordance with various examples. The cross-section of FIG. **3**A is taken along line **390** and the cross-section of FIG. **3**B is taken along line **392**, as shown in FIG. **3**C. The MEMS device **350** of FIGS. **3**A-**3**D is substantially similar to the MEMS device **250** of FIGS. **2**A-**2**D, except that the structure **216** in the MEMS device **350** of FIGS. **3**A-**3**D has a more specific composition. In particular, the structure **216** includes an interposer **300**, such as a semiconductor (e.g., silicon, gallium nitride) interposer, that is positioned in between and contacting the cap **218** and a metal stack **302**. The metal stack **302** may include multiple different metals, multiple metal alloys, multiple oxide layers, glass, semiconductors, and other materials that together facilitate the formation and preservation of a seal, for example a hermetic seal, protecting the cavity **220**. Example materials in the metal stack **302** include copper, nickel, etc. Other materials not specifically listed are also contemplated and included in the scope of this disclosure. Different layers in the metal stack **302** may have different shapes and physical dimensions. In some examples, the MEMS device **350** may include a single metal stack, and in other examples, the MEMS device **350** may include multiple metal stacks. Any of a variety of configurations of the metal stack(s) and semiconductor interposer(s) are contemplated and included in the scope of this disclosure.

[0036] FIG. **3**E is a cross-sectional view of an example MEMS device **352**, which is an example of MEMS device **104** and is identical to the example MEMS device **350** of FIGS. **3**A-**3**D, except that an epoxy **215** is extended to one or more edges of the perimeter of the substrate **200**, as in FIG. **2**E. FIG. **3**F is a top-down view of the MEMS device **352** of FIG. **3**E, in accordance with various examples. FIG. **3**F shows where the cross-section of FIG. **3**E is taken, along line **354**. FIG. **3**G is a perspective view of the MEMS device **352** of FIGS. **3**E and **3**F, in accordance with various examples.

[0037] As described above, in some examples of the MEMS device **104** (e.g., MEMS devices **252** and **352**), the epoxy **215** extends to one or more edges of a perimeter of the substrate **200** (e.g., as shown in FIGS. **2**E-**2**G, and **3**E-**3**G). In other examples of the MEMS device **104** (e.g., MEMS devices **250** and **350**), the epoxy **214** does not extend to the edges of the perimeter of the substrate **200** (e.g., as shown in FIGS. **2**A-**2**D and **3**A-**3**D). Techniques for manufacturing each of these two types of MEMS devices are now described. Specifically, FIG. **4** is a flow diagram of a method **400** for manufacturing a MEMS device **104** (e.g., MEMS device **252** or **352**) with epoxy **215** extending to the edges of the substrate **200** perimeter, and FIGS. **5**A**1**-**5**E**3** are a process flow for manufacturing the MEMS device **104** (e.g., MEMS device **252** or **352**) with epoxy **215** extending to the edges of the substrate **200** perimeter. Thus, FIGS. **4** and **5**A**1**-**5**E**3** are now described in parallel. Further, FIG. **6** is a flow diagram of a method **600** for manufacturing a MEMS device **104** (e.g., MEMS device **250** or **350**) with epoxy **214** that does not extend to the edges of the substrate **200** perimeter, and FIGS. **7**A**1**-**7**E**3** are a process flow for manufacturing the MEMS device **104** (e.g., MEMS device **250** or **350**) with epoxy **214** that does not extend to the edges of the substrate **200** perimeter. Thus, FIGS. **6** and **7**A**1**-**7**E**3** are described in parallel, further below.

[0038] Referring now to FIGS. **4** and **5**A**1**-**5**E**3**, the method **400** (the steps of which may be performed in any suitable order) includes coupling a first device to a first substrate of a substrate panel (**402**). The first device includes a first semiconductor die (**402**). The method **400** includes wirebonding the first device to the first substrate (**403**). The method **400** also includes coupling a first structure to the first device such that the first structure extends approximately orthogonally away from and around the first semiconductor die, where the first structure includes a first, optional semiconductor interposer, a first metal stack, and a cap (**404**).The method **400** also includes coupling a second device to a second substrate of the substrate panel (**405**). The second device includes a second semiconductor die (**405**). The method **400** includes wirebonding the second device to the second substrate (**406**). The method **400** also includes coupling a second structure to the second device such that the second structure extends approximately orthogonally away from and around the second semiconductor die, where the second structure includes a second, optional semiconductor interposer, a second metal stack, and a cap (**407**). The first and second substrates are coupled to each other (**407**). FIG. **5**A**1** is a top-down view of an example substrate panel **500** that includes four regions, which will later be singulated to become four substrates **200**, joined along perforations **502**. FIG. **5**A**2** is a profile view of the substrate panel **500**, and FIG. **5**A**3** is another profile view of the substrate panel **500**. Although the example substrate panel **500** includes four substrates **200**, other example substrate panels **500** may include more or fewer substrates **200**. In examples, the substrate panel **500** is scored with perforations **502** between the substrates **200** to facilitate subsequent singulation of the substrate panel **500** into individual substrates **200**. The substrates **200** include bond leads **208**. Each substrate **200** includes a device, which includes at least a semiconductor die **202** and a structure **216**. FIG. **5**B**1** is a top-down view of the substrate panel **500** of FIG. **5**A**1**, except that multiple devices are wirebonded to the substrates **200** of the substrate panel **500**. FIGS. **5**B**2** and **5**B**3** are profile views of the structure of FIG. **5**B**1**, in accordance with various examples. These devices are similar to the MEMS devices **104** described herein, except that the substrates **200** and epoxies **215** are not yet parts of the devices. (After the devices are coupled to the substrates **200** and epoxies **215** have been deposited, it may be said that multiple MEMS devices **104** have been formed, but for the sake of consistency in nomenclature, the devices are referred to as “MEMS devices” instead of “MEMS devices **104**” until the substrates **200** and epoxies **215** are added, at which point the devices may be considered “MEMS devices **104**.”) Such coupling may include coupling semiconductor dies of the devices to the substrates **200** (e.g., using die attach layers), wirebonding bond pads of the devices to the bond leads **208** of the substrates **200**, and so on.

[0039] The method **400** includes depositing an epoxy (e.g., such as by dispensing the epoxy using a syringe- or cartridge-based system and a dispensing tip or nozzle) contacting first and second surfaces of the first structure and contacting third and fourth surfaces of the second structure (**408**). The method **400** includes depositing the epoxy contacting fifth and sixth surfaces of the first structure and contacting seventh and eighth surfaces of the second structure (**410**). The method **400** includes depositing the epoxy in a gap between the sixth surface of the first structure and the seventh surface of the second structure (**412**). FIG. **5**C**1** is a top-down view of the structure of FIG. **5**B**1**, except that epoxy (e.g., mold compound) is deposited contacting first and second surfaces of the structure **216** shown in the top-left of FIG. **5**C**1** and contacting third and fourth surfaces of the structure **216** shown in the bottom-left of FIG. **5**C**1**. Similarly, epoxy is deposited contacting first and second surfaces of the structure **216** shown in the top-right of FIG. **5**C**1** and contacting third and fourth surfaces of the structure **216** shown in the bottom-right of FIG. **5**C**1**. FIGS. **5**C**2** and **5**C**3** are profile views of the structure of FIG. **5**C**1**, in accordance with various examples. FIG. **5**D**1** is a top-down view of the structure of FIG. **5**C**1**, except that epoxy (e.g., mold compound) is deposited contacting fifth and sixth surfaces of the structure **216** on the top-left of FIG. **5**D**1** and contacting seventh and eighth surfaces of the structure **216** on the bottom-left. Similarly, epoxy is deposited contacting fifth and sixth surfaces of the structure **216** on the top-right and contacting seventh and eighth surfaces of the structure **216** on the bottom-right. Epoxy is also deposited in the gap between the sixth surface of the structure **216** on the top-left and the seventh surface of the structure **216** on the bottom-left, and likewise between the sixth surface of the structure **216** on the top-right and the seventh surface of the structure **216** on the bottom-right, as numerals **506** indicate. FIGS. **5**D**2** and **5**D**3** are profile views of the structure of FIG. **5**D**1**, in accordance with various examples.

[0040] Deposition of epoxy in the areas indicated by numerals **506** and subsequent singulation along the perforations **502** results in a MEMS device in which the epoxy extends to the edge of the perimeter of the substrate **200**. For example, as FIG. **5**D**1** shows, epoxy is deposited on the top-left substrate **200** of the substrate panel **500** and on the bottom-left substrate **200** of the substrate panel **500** without a break or interruption in the epoxy (as FIG. **5**D**3** shows). Thus, later singulation along the perforation **502** that is between the top-left and bottom-left substrates **200** will result in a MEMS device in which the epoxy extends to the edge of the top-left substrate **200** that bordered the bottom-left substrate **200** prior to singulation, and, likewise, in a MEMS device in which the epoxy extends to the edge of the bottom-left substrate **200** that bordered the top-left substrate **200** prior to singulation. The epoxy may also extend to any other edges of a substrate **200** where the epoxy was similarly applied as between the top-left and bottom-left substrate **200** of the substrate panel **500**. (Although the substrate panel **500** may have any number of substrates **200** in the horizontal plane, the terms “top-left substrate,” “bottom-left substrate,” etc. used with reference to FIG. **5**D**1** refer to the four substrates **200** expressly shown in FIG. **5**D**1**.) The epoxy deposition technique shown in FIGS. **5**D**1** and **5**D**3** may be advantageous and useful in any of a variety of contexts, such as when the available space between consecutively adjacent semiconductor dies is minimal, or when faster manufacturing speed is especially desirable. The epoxy deposition technique shown in FIGS. **5**D**1** and **5**D**3** may be applied to selectively extend the epoxy to one or more edges of a substrate **200**. In some instances, it may be disadvantageous, or of modest benefit, to extend the epoxy to the substrate edges in this manner. For example, in FIG. **5**D**1**, the epoxy does not extend to the perforation **502** between the top-left substrate **200** and the top-right substrate **200**, resulting in the cross-sectional view provided in FIG. **5**D**2**. In this approach, epoxy is applied on the surface **242**, just outside the perimeter of the shelf **244**. Leaving a perforation **502** free of epoxy is advantageous because the substrate panel **500** may be singulated along the perforation **502** by applying pressure to break the substrate panel **500** rather than by using a saw to cut the substrate panel **500**.

[0041] The method **400** includes singulating the first and second substrates from each other by cutting through the epoxy in the gap and through a portion of the substrate panel that is in vertical alignment with the gap (**414**). In some examples, the method **400** may include cutting through the epoxy in the gap and scribing and breaking the substrate panel in the area that is in vertical alignment with the gap. FIG. **5**E**1** is a top-down view of the structure of FIG. **5**D**1**, except that the substrate panel **500** is being singulated as numeral **510** depicts, in accordance with various examples. FIGS. **5**E**2** and **5**E**3** are profile views of the structure of FIG. **5**E**1**, in accordance with various examples. As FIGS. **5**E**1**-**5**E**3** show, the epoxy **215** contacts all four sides of the structure **216**, thereby sealing and protecting the various components of the structure **216** from damage (e.g., oxidation) that would otherwise occur by exposure to the environment. In addition to contacting all four sides of the structure **216**, the four corners of the structure **216** are likewise covered by the epoxy **215**, thus forming a barrier that circumscribes the entirety of the structure **216**. The structures that result from performance of the method **400** and the process flow in FIGS. **5**A**1**-**5**E**3** are shown in FIGS. **2**E, **2**F, and **2**G.

[0042] Referring now to FIGS. **6** and **7**A**1**-**7**E**3**, a method **600** (the steps of which may be performed in any suitable order) for manufacturing MEMS devices **104** (e.g., MEMS devices **250** and **350**) includes coupling a first device to a first substrate of a substrate panel (**602**). The first device includes a first semiconductor die (**602**). The method **600** includes wirebonding the first device to the first substrate (**603**). The method **600** also includes coupling a first structure to the first device such that the first structure extends approximately orthogonally away from and around the first semiconductor die, where the first structure includes a first, optional semiconductor interposer, a first metal stack, and a cap (**604**). The method **600** includes coupling a second device to a second substrate of the substrate panel (**605**). The second device includes a second semiconductor die (**605**). The method **600** includes wirebonding the second device to the second substrate (**606**). The method **600** also includes coupling a second structure to the second device such that the second structure extends approximately orthogonally away from and around the second semiconductor die, where the second structure includes a second, optional semiconductor interposer, a second metal stack, and a cap (**607**). The first and second substrates are coupled to each other (**607**). FIG. **7**A**1** is a top-down view of an example substrate panel **500** having multiple substrates **200** separated from each other by perforations **502**. Devices are coupled to the substrates **200**, and these devices include at least semiconductor dies **202** and structures **216**. FIGS. **7**A**2** and **7**A**3** are profile views of the structure of FIG. **7**A**1**, in accordance with various examples. FIG. **7**B**1** is a top-down view of the structure of FIG. **7**A**1**, except that the bond pads of the MEMS devices are coupled to the bond pads of the substrates **200** by way of wirebonding. FIGS. **7**B**2** and **7**B**3** are profile views of the structure of FIG. **7**B**1**, in accordance with various examples.

[0043] The method **600** includes depositing an epoxy contacting first and second surfaces of the first structure and contacting third and fourth surfaces of the second structure (**608**). The method **600** also includes depositing the epoxy contacting fifth and sixth surfaces of the first structure and contacting seventh and eighth surfaces of the second structure (**610**). FIG. **7**C**1** is a top-down view of the structure of FIG. **7**B**1**, except that epoxy **700** (e.g., mold compound) has been deposited contacting first and second surfaces of the structures **216** on the top-left and top-right and contacting third and fourth surfaces of the structures **216** on the bottom-left and bottom-right. FIGS. **7**C**2** and **7**C**3** are profile views of the structure of FIG. **7**C**1**, in accordance with various examples. FIG. **7**D**1** is a top-down view of the structure of FIG. **7**C**1**, except that epoxy **704** (e.g., mold compound) has been deposited contacting the fifth and sixth surfaces of the structure **216** on the top-left and top-right and on seventh and eighth surfaces of the structure **216** on the bottom-left and bottom-right. FIGS. **7**D**2** and **7**D**3** are profile views of the structure of FIG. **7**D**1**, in accordance with various examples. As opposed to the examples of FIGS. **5**D**1**-**5**D**3**, in which epoxy was extended to at least some substrate **200** edges, in the examples of FIGS. **7**D**1**-**7**D**3**, the epoxy is not extended to any of the substrate **200** edges, which may be advantageous because a saw is not needed to singulate the substrate panel **500** along the perforations **502**. Specifically, numerals **750**, **752**, **754**, and **756** indicate gaps, i.e., the absence of epoxy between each pair of the four semiconductor dies **202** shown in FIG. **7**D**1**.

[0044] The method **600** includes singulating (e.g., cutting or scribing and breaking) the first and second substrates from each other (**612**). FIG. **7**E**1** is a top-down view of the structure of FIG. **7**D**1**, except that the substrate panel **500** is being singulated into individual substrates **200**, as numeral **702** indicates. The singulation produces individual MEMS devices **104** (e.g., MEMS devices **250** or **350**). FIGS. **7**E**1** and **7**E**2** are profile views of the structure of FIG. **7**E**1**, in accordance with various examples. FIGS. **2**A-**2**D show cross-sectional, cross-sectional, top-down, and perspective views, respectively, of the completed MEMS devices **104** (e.g., MEMS devices **250** or **350**) formed by the performance of method **600** and the process flow of FIGS. **7**A**1**-**7**E**3**, in accordance with various examples. As FIGS. **7**E**1**-**7**E**3** show, the epoxy **214** contacts all four sides of the structure **216**, thereby sealing and protecting the various components of the structure **216** from damage (e.g., oxidation) that would otherwise occur by exposure to the environment. In addition to contacting all four sides of the structure **216**, the four corners of the structure **216** are likewise covered by the epoxy **214**, thus forming a barrier that circumscribes the entirety of the structure **216**.

[0045] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device

B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0046] A device that is described herein as including certain components may instead be coupled to those components to form the described device. For example, a structure described as including one or more elements (such as one or more processors and/or controllers) may instead include one or more elements within a single physical device (e.g., a display device) and may be coupled to at least some of the elements to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0047] In this description, unless otherwise stated, "about," "approximately" or "substantially" preceding a parameter means being within +/−10 percent of that parameter. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.

Claims

1. A micro-electromechanical (MEMS) device, comprising: a substrate; a semiconductor die coupled to the substrate and including circuitry formed therein, the semiconductor die also including a bond pad coupled to the circuitry; a structure extending away from the semiconductor die and having four sides, the structure comprising a corrodible material; and an epoxy contacting outer surfaces of the four sides of the structure and over the corrodible material.

2. The MEMS device of claim 1, further comprising a first bond wire coupling the semiconductor die to a first side of the substrate and a second bond wire coupling the semiconductor die to a second side of the substrate, the second side of the substrate opposite the first side of the substrate, the epoxy over the first wire bond and the second wire bond.

3. The MEMS device of claim 1, wherein the MEMS device further comprises a cap coupled to the semiconductor die by way of the structure to form a sealed cavity between the cap, the structure, and the semiconductor die.

4. The MEMS device of claim 3, wherein the cap is a glass member, and wherein the epoxy does not contact a surface of the glass member that faces away from the semiconductor die.

5. The MEMS device of claim 1, wherein the structure comprises a metal stack coupled to a semiconductor interposer.

6. The MEMS device of claim 1, wherein the epoxy extends to an edge of the substrate.

7. The MEMS device of claim 1, wherein a gap separates the epoxy from an edge of the substrate.

8. A micro-electromechanical (MEMS) device, comprising: a substrate including a bond lead; a semiconductor die coupled to the substrate and including a bond pad coupled to the bond lead by way of a bond wire, the semiconductor die having a device side; a four-sided structure comprising a metal stack, the four-sided structure coupled to the device side of the semiconductor die and to a glass member and forming a sealed cavity between the semiconductor die and the glass member; and an epoxy contacting the metal stack on the four sides of the four-sided structure and over the bond wire.

9. The MEMS device of claim 8, wherein the metal stack comprises multiple different metals and multiple oxide layers, and wherein the multiple different metals comprise copper and nickel.

10. The MEMS device of claim 8, wherein the epoxy extends to an edge of the substrate.

11. The MEMS device of claim 8, wherein a gap separates the epoxy from an edge of the substrate.

12. The MEMS device of claim 8, wherein the metal stack comprises at least one of glass, oxide layers, and alloys.

13. The MEMS device of claim 8, wherein the epoxy has a coefficient of thermal expansion below 25.

14. The MEMS device of claim 8, wherein the four-sided structure comprises a semiconductor interposer coupled to the metal stack.

15. The MEMS device of claim 8, wherein the epoxy does not contact any part of a top surface of the glass member, the top surface of the glass member facing away from the semiconductor die.

16. A method of manufacturing a micro-electromechanical (MEMS) device, comprising: attaching a first semiconductor die to a substrate panel; attaching a first structure to the substrate panel around the first semiconductor die, the first structure comprising a first metal stack; attaching a second semiconductor die to the substrate panel; attaching a second structure to the substrate panel around the second semiconductor die, the second structure comprising a second metal stack; depositing an epoxy contacting first and second surfaces of the first structure and contacting third and fourth surfaces of the second structure and in a gap between the second surface of the first structure and the third surface of the second structure; and singulating the substrate panel comprising cutting through epoxy in the gap and through a portion of the substrate panel.

17. The method of claim 16, wherein the epoxy is composed of at least 80% silica.

18. The method of claim 16, wherein the epoxy has a coefficient of thermal expansion below 25.

19. The method of claim 16, wherein each of the first and second structures includes a semiconductor interposer.

20. The method of claim 16, further comprising attaching a glass member to the first semiconductor die by way of the first structure, and wherein a top surface of the glass member facing away from the first semiconductor die is not covered by the epoxy.
